

Ankit Choudhary

Software Engineer II | Backend & Distributed Systems | LLM / Conversational AI

Software Engineer with **5+ years of experience** building scalable full-stack products, distributed systems, and event-driven microservices, currently building **LLM-powered conversational AI agents (voice and SMS)** for automotive dealerships. Strong expertise in **Node.js, React.js, Next.js, Kafka, MongoDB, Redis, AWS Step Functions** and image/video processing pipelines. Proven track record of cutting processing time, reducing TAT, building orchestration layers, and leading engineering teams in fast-paced startups.

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TECHNICAL SKILLS

Languages	JavaScript (ES6+), Python, SQL
Backend	Node.js, Express.js, Sails.js, REST APIs, Microservices
Messaging & Streaming	Kafka, AWS SQS, Bull Queue, WebSockets, Event-Driven Architecture
AI & LLM	Large Language Models (LLMs), Conversational AI, Voice AI (LiveKit), Twilio SMS, LLM-based Testing
Frontend	React.js, Next.js, Redux, Tailwind CSS, Material UI, Bootstrap
Databases	MongoDB, MySQL, Redis, ClickHouse, DynamoDB
Cloud (AWS)	AWS Step Functions, SQS, EventBridge Scheduler, DynamoDB
Core Concepts	System Design (HLD/LLD), Data Structures & Algorithms, Scalability, Observability

KEY HIGHLIGHTS

- **66% faster AI image processing** (90 → 30 min) at **~1M concurrent images**
- **~60% faster** multi-product processing after **monolith → microservices** migration
- **~90% fewer webhook events** (10 → 1 per SKU) with event-driven redesign
- Building **Vini**, an **LLM-powered voice & SMS agent** for automotive dealerships
- **~85% fewer pipeline failures** (7% → <1%) via observability & retries
- **Led a team of 6 engineers** and built **Acadza's first production platform**

WORK EXPERIENCE

Software Engineer II

Spyne

04/2025 - Present *GURUGRAM*

AI-powered visual merchandising and conversational AI platform for automotive dealerships and e-commerce.

- Building **Vini**, an **LLM-powered conversational agent** that handles automotive dealer **sales and service conversations over voice (LiveKit) and SMS (Twilio)**, with real-time integration into dealer **inventory and CRM systems**.
- Designed the **automated SMS follow-up system** for inconclusive sales calls: eligibility gates, **EventBridge Scheduler**-based debounce, a dedicated Eligibility API, **DynamoDB/Redis** state store, and a **Kafka → ClickHouse** audit trail.
- Built an **LLM-based automated testing setup**, significantly reducing manual testing effort for releases.
- Architected **core AI image-processing orchestration services** using **AWS Step Functions**, achieving a **66% reduction in processing time (90 → 30 minutes)** while scaling to **~1M images processed concurrently** (10K images × 100 enterprises).
- Led migration from a **monolithic architecture to microservices**, reducing multi-product processing time from **45–60 minutes to ~20 minutes (~60% improvement)** by enabling parallel processing of **images, videos, and 360° media pipelines**.
- Designed an **event-driven webhook processing platform**, cutting webhook volume by **~90% (10 → 1 per SKU)**, reducing client server load and eliminating duplicate event issues.
- Built an **auto-SKU assignment system** driven by enterprise **TAT/SLA priorities**, automating manual QC allocation and reducing delivery time by **~53% (15 → 7 minutes per SKU)**.
- Enhanced **observability and retry strategies** in distributed image-processing workflows, cutting pipeline failure rates by **~85% (7% → <1%)**.
- Developed **reusable internal npm packages** for shared dashboards, unifying multiple platform implementations into a **single standardized system** and reducing duplicated development effort.

WORK EXPERIENCE

Software Engineer
Spyne

05/2023 - 03/2025 *GURUGRAM*

AI-powered image and video transformation platform for automotive and e-commerce.

- Designed a **multi-assignment QC system**, allowing users to assign multiple SKUs concurrently instead of one at a time, reducing **per-SKU QC time by 75% (2 min → 30 sec)**.
- Developed real-time **Image Enhancer** and **Virtual Studio** tools supporting multiple backgrounds and studio transformations.
- Designed an **AI-powered broken-frame video recovery system** that generates 360° spin image sequences.
- Developed a **video template engine** for automated marketing video generation using dynamic user assets.
- Built an **orchestration layer** to generate multiple product categories from a single media source.
- Built **centralized internal dashboards** unifying operational and product metrics.

Software Engineer
Acadza Technology Pvt Ltd

01/2021 - 04/2023 *MUMBAI*

EdTech startup focused on scalable digital learning products.

- **Built and led a team of 6 engineers**, mentoring them on delivery, debugging, and scalable frontend practices.
- Developed the company's **first production-grade web platform from scratch**.
- Created an **internal frontend framework** using **Next.js, Redux, Context API, custom hooks, and authentication modules**, significantly improving development velocity.
- Integrated **Razorpay payment workflows** with secure transaction handling.
- Implemented **WebSocket-based real-time dashboards** and charting systems for analytics use cases.

EDUCATION

B.Tech, Mechanical Engineering
Swami Vivekanand Institute of Engineering & Technology, Punjab

2014 - 2018

LANGUAGES

English
Professional Working Proficiency

Hindi
Native or Bilingual Proficiency